MPC-MAP Assignment No. 1 – Report

Week 2: Uncertainty

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Task 2 – Sensor Uncertainty

Static measurements were collected at position $[17, 5, 0]$ on `mixed_1` over 150 iterations.

LiDAR.

Ch.	1	2	3	4	5	6	7	8
σ [m]	0.054	0.051	0.052	0.048	0.051	0.049	0.049	0.053

All channels have similar $\sigma \approx 0.05$ m.

GNSS. $\sigma_X = 0.492$ m, $\sigma_Y = 0.512$ m, about $10\times$ noisier than LiDAR.

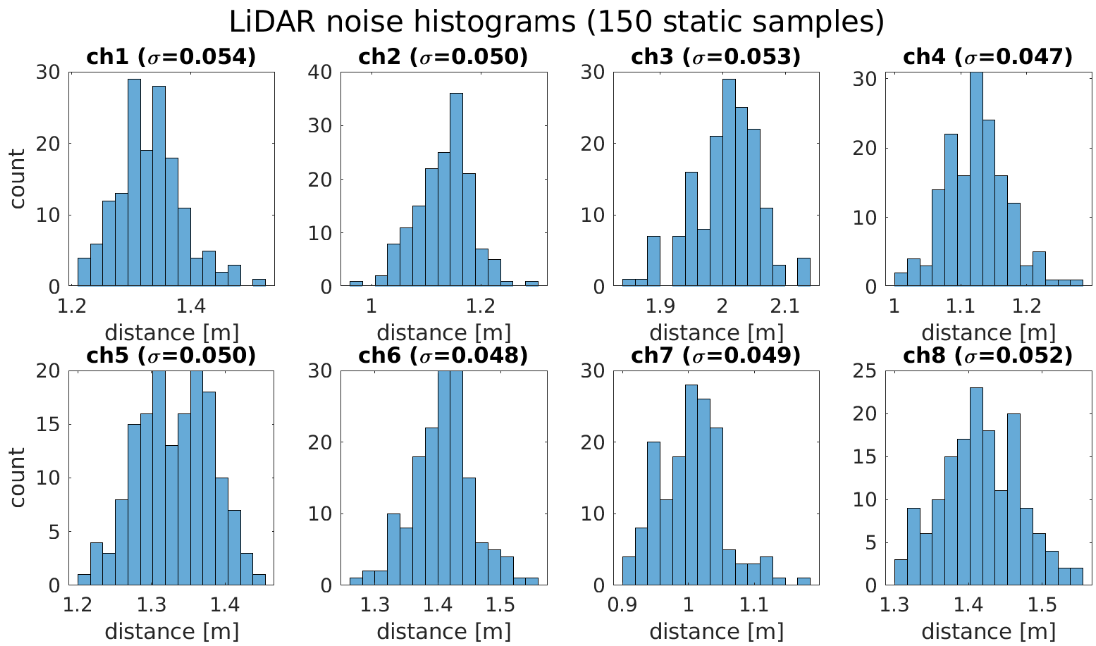

Figure 1: LiDAR noise histograms (150 samples).

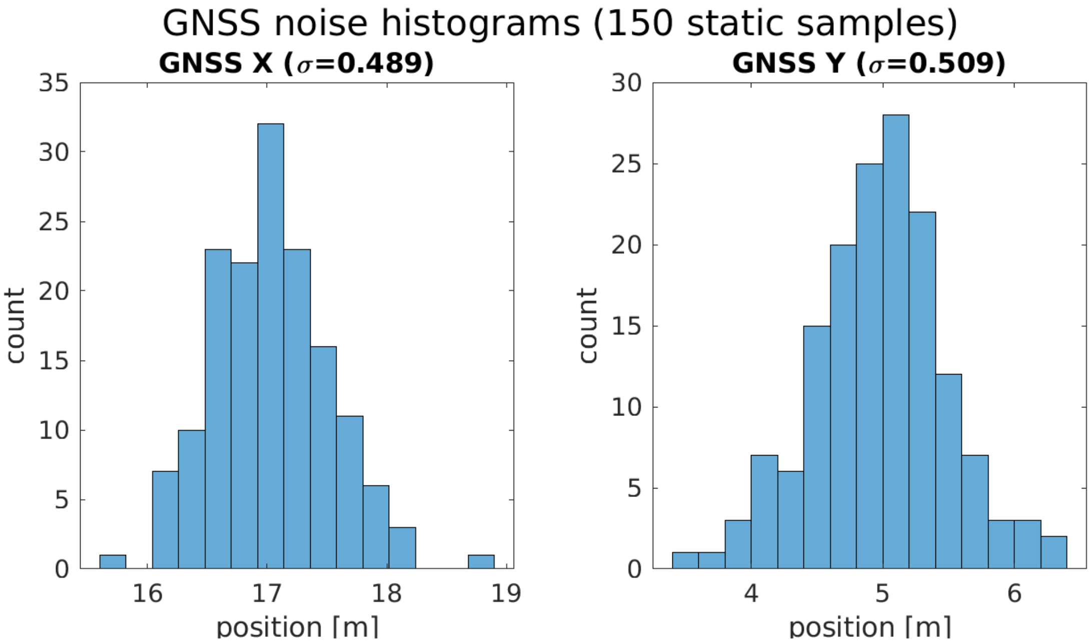

Figure 2: GNSS noise histograms (150 samples).

Task 3 – Covariance Matrix

Covariance matrices computed with `cov()`.

GNSS covariance (2×2).

$$\mathbf{C}_{\mathrm{GNSS}} = \begin{pmatrix} 0.242 & -0.008 \\ -0.008 & 0.263 \end{pmatrix}$$

The diagonal values match σ^2 from Task 2. The off-diagonal is close to zero, so X and Y noise is nearly uncorrelated.

LiDAR covariance (8×8).

$$\mathbf{C}_{\mathrm{LiDAR}} = \begin{pmatrix} 0.0029 & 0.0000 & 0.0001 & 0.0000 & 0.0000 & 0.0001 & 0.0000 & 0.0000 \\ 0.0000 & 0.0026 & 0.0000 & 0.0000 & 0.0001 & 0.0000 & 0.0000 & 0.0001 \\ 0.0001 & 0.0000 & 0.0027 & 0.0000 & 0.0000 & 0.0000 & 0.0001 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0023 & 0.0000 & 0.0001 & 0.0000 & 0.0000 \\ 0.0000 & 0.0001 & 0.0000 & 0.0000 & 0.0026 & 0.0000 & 0.0000 & 0.0000 \\ 0.0001 & 0.0000 & 0.0000 & 0.0001 & 0.0000 & 0.0024 & 0.0000 & 0.0001 \\ 0.0000 & 0.0000 & 0.0001 & 0.0000 & 0.0000 & 0.0000 & 0.0024 & 0.0000 \\ 0.0000 & 0.0001 & 0.0000 & 0.0000 & 0.0000 & 0.0001 & 0.0000 & 0.0028 \end{pmatrix}$$

The diagonal entries match σ^2 from Task 2 and the off-diagonal values are close to zero, so the channels are independent.

Task 4 – Normal Distribution

Implemented `norm_pdf(x, mu, sigma)`:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

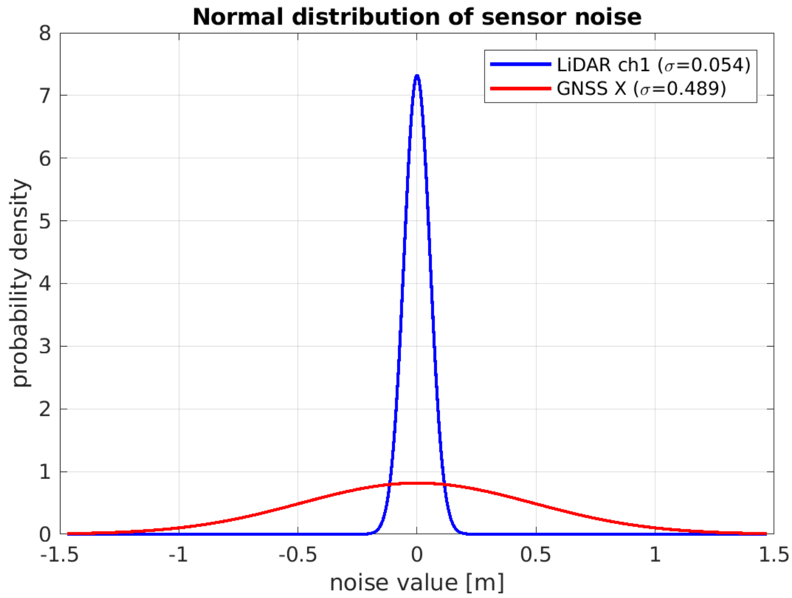

Figure 3: Normal PDFs for LiDAR (narrow) and GNSS (wide) noise.

Task 5 – Motion Uncertainty

An open-loop path was programmed in `plan_motion.m` to navigate to the goal on `indoor_1`. Because of motion noise, only about 20% of attempts succeed. Small errors in the wheel velocities cause the robot to drift from the planned path, and without sensor feedback there is no way to correct them.

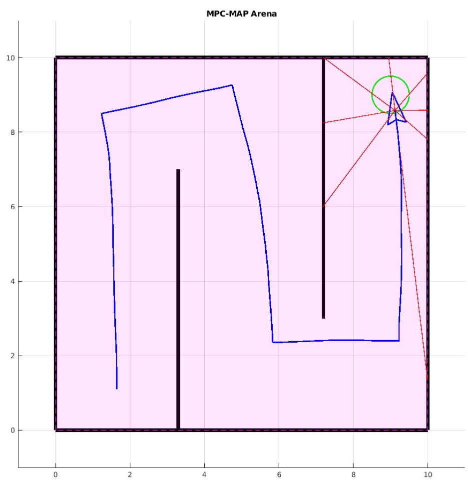

Figure 4: Successful open-loop run on `indoor_1`.